# Tables

1 Consider the function $f(x) = \dfrac{1}{7 - x}$.

a Complete the first table of values, in which $x < 7$

b Complete the second table of values, in which $x > 7$

c Find the limit of $f(x)$ as the value of $x$ approaches 7.

| $x$ | $5$ | $6$ | $6.9$ | $6.99$ |
|---|---|---|---|---|
| $f(x)$ | | | | |

| $x$ | $9$ | $8$ | $7.1$ | $7.01$ |
|---|---|---|---|---|
| $f(x)$ | | | | |

2 Consider the function $f(x) = 5x^2 + 1$:

a Complete the table to find the exact values of $f(x)$ as $x$ gets closer and closer to $2$ from the left, and closer and closer to $2$ from the right:

| $x$ | $1.9$ | $1.99$ | $1.999$ | $2.001$ | $2.01$ | $2.1$ |
|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | |

b Find the value of $\lim_{x \to 2} \left(5x^2 + 1\right)$.

3 Consider the function $f(x) = \dfrac{2 - x}{x^2 + 2}$:

a Complete the table to find the values of $f(x)$ as $x$ gets closer and closer to $0$ from the left, and closer and closer to $0$ from the right. Round your answers to four decimal places.

| $x$ | $-0.1$ | $-0.01$ | $-0.001$ | $0.001$ | $0.01$ | $0.1$ |
|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | |

b Find the value of $\lim_{x \to 0} \left(\dfrac{2 - x}{x^2 + 2}\right)$.

4 Consider the function $f(x) = \dfrac{x^2 - 4x}{x - 4}$.

a Complete the table to find the values of $f(x)$ as $x$ gets closer and closer to $4$ from the left, and closer and closer to $4$ from the right:

| $x$ | $3.9$ | $3.99$ | $3.999$ | $4.001$ | $4.01$ | $4.1$ |
|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | |

b $\left(x^2 - 4x\right.$

5 Consider the function $f(x) = \frac{x^3 + x + 2}{x + 1}$.

a Complete the following table:

| $x$ | $-1.1$ | $-1.01$ | $-1.001$ | $-0.999$ | $-0.99$ | $-0.9$ |
|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | |

b Find $\lim_{x \to -1} f(x)$.

6 Consider the function $f(x) = \frac{\sqrt{x} + 4}{x - 5}$ :

a Complete the following table, rounding all values to two decimal places:

| $x$ | $4.9$ | $4.99$ | $4.999$ | $5.001$ | $5.01$ | $5.1$ |
|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | |

b Does $\lim_{x \to 5} f(x)$ exist? Explain your answer.

7 Consider the limit: $\lim_{x \to -5} \left( \frac{x^2 + 4}{x + 5} \right)$.

a Complete the table.

| $x$ | $-5.1$ | $-5.01$ | $-5.001$ | $-5$ | $-4.999$ | $-4.99$ | $-4.9$ |
|---|---|---|---|---|---|---|---|
| $\frac{x^2 + 4}{x + 5}$ | | | | – | | | |

b Does the above limit exist? Explain your answer.

8 Consider the limit: $\lim_{x \to 3} \left( \frac{e^{x-3} + x - 4}{x - 3} \right)$.

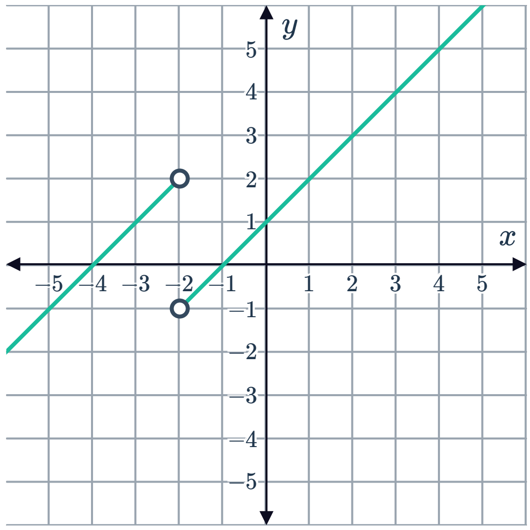
a Complete the following table:

| $x$ | $2.9$ | $2.99$ | $2.999$ | $3$ | $3.001$ | $3.01$ | $3.1$ |
|---|---|---|---|---|---|---|---|
| $\left( \frac{e^{x-3} + x - 4}{x - 3} \right)$ | | | | – | | | |

b Does the above limit exist? Explain your answer.

## Graphs

**9** Consider the graph of the piecewise function $g(x)$:

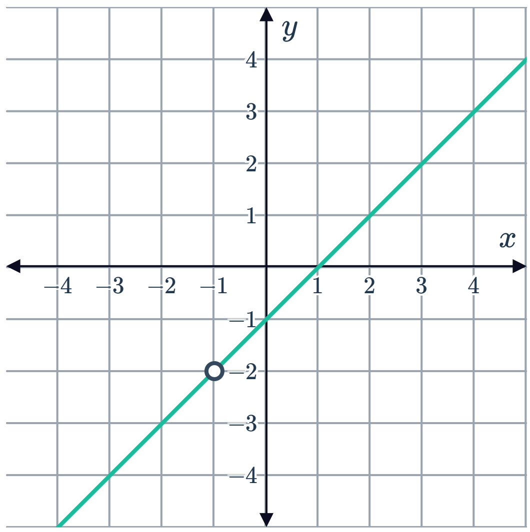



a If we start at $-4$ and move along $g(x)$ to the right towards $x=-2$, what $y$-value do we approach?

b If we start at $0$ and move along $g(x)$ to the left towards $x=-2$, what $y$-value do we approach?

c Does the $\lim_{x\to-2} g(x)$ exist? Explain your answer.

**10** Consider the graph of the function $f(x)=\dfrac{x^2-1}{x+1}$:



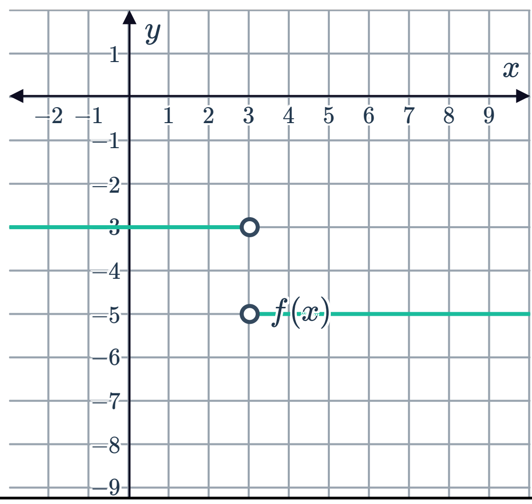

a If we start at $x=-3$ and move along the function to the right towards $x=-1$, what $y$-value do we approach?

b If we start at $x=1$ and move along the function to the left towards $x=-1$, what $y$-value do we approach?

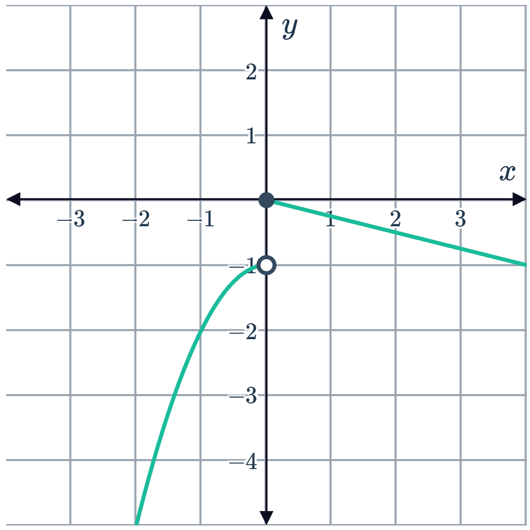
c Describe the behaviour of $f(x)=\dfrac{x^2-1}{x+1}$ as $x$ approaches $-1$.

d Write part (c) using limit notation.

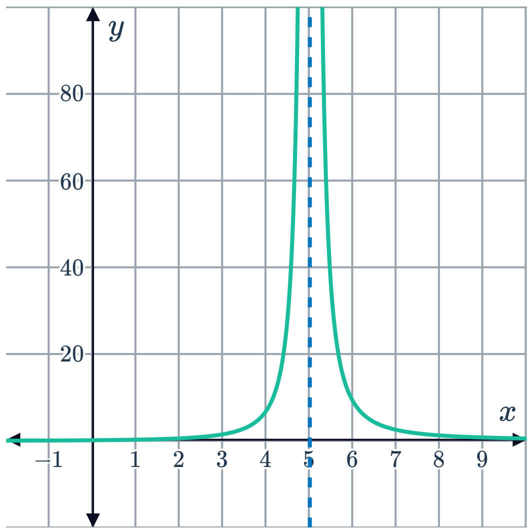
**11** Consider the graph of $f(x)$:
Does the limit $\lim_{x\to3} f(x)$ exist? Explain your answer.



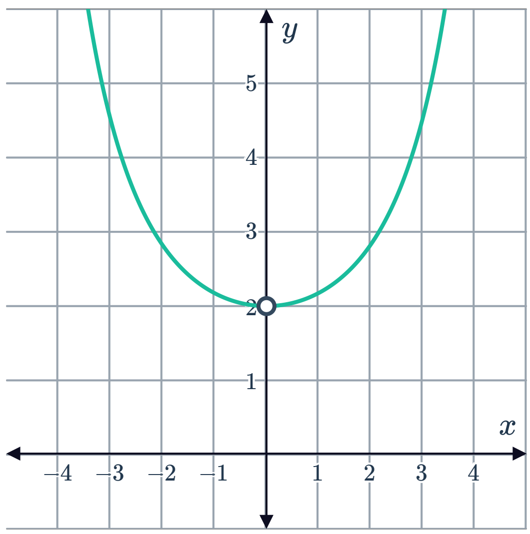
**12** Consider the graph of $f(x)$:
Does the limit $\lim_{x \to 0} f(x)$ exist? Explain your answer.



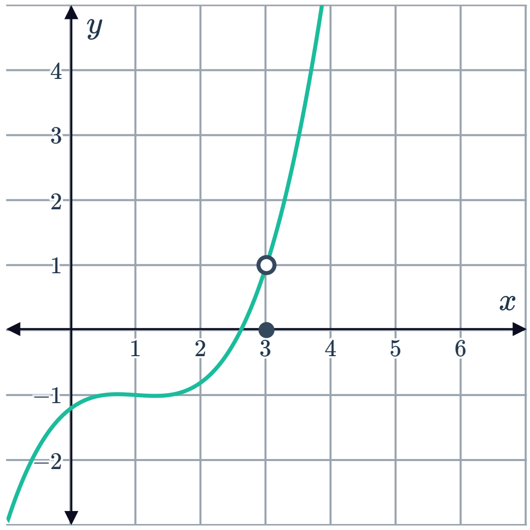

**13** Consider the graph of $y = \dfrac{x+3}{(x-5)^2}$:
Does the limit $\lim_{x \to 5} \left( \dfrac{x+3}{(x-5)^2} \right)$ exist?
Explain your answer.



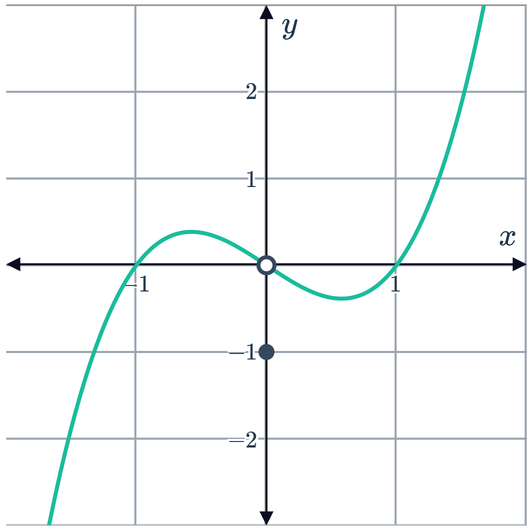

**14** Consider the graph of $y = \dfrac{x^2}{1 - \cos x}$:
Does the limit $\lim_{x \to 0} \left( \dfrac{x^2}{1 - \cos x} \right)$ exist?
Explain your answer.



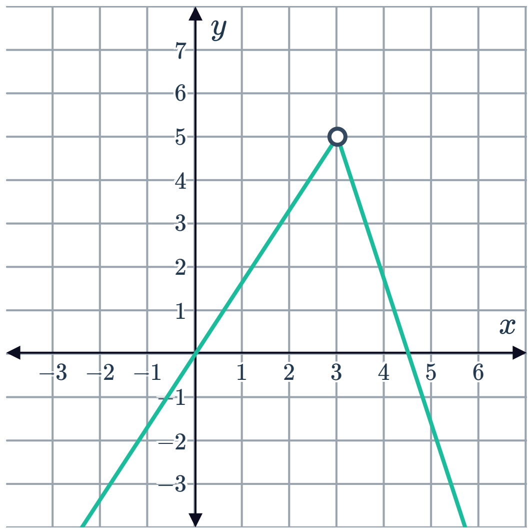
**15** Consider the graph of the function $f(x)$:
Find the value of $\lim_{x\to 3} f(x)$.



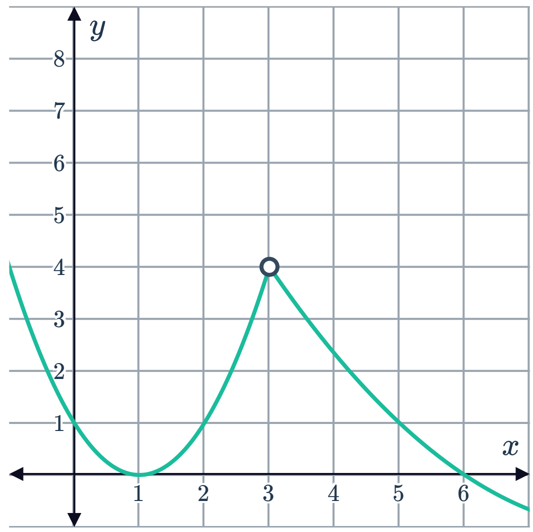

**16** Consider the graph of the function $f(x)$:
Find the value of $\lim_{x\to 0} f(x)$.



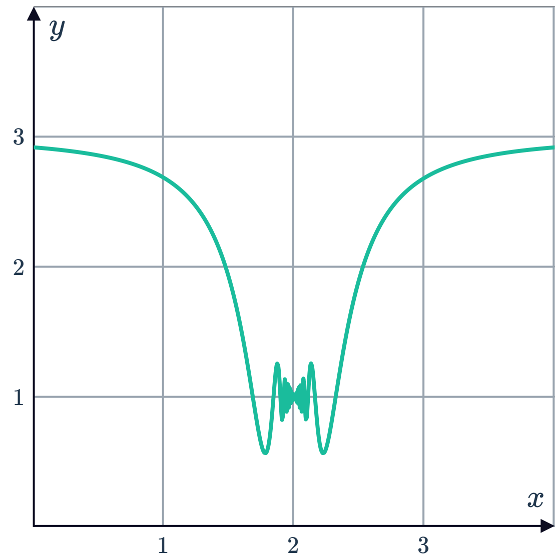

**17** Consider the graph of the function $f(x)$:
Find the value of $\lim_{x\to 3} f(x)$.



**18** Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 3} f(x)$.




**19** Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 2} f(x)$.




**20** Consider the graph of the function

$f(x) = \dfrac{1}{x+4}$:

Find the value of:

**a** $\lim_{x \to -4^+} \left(\dfrac{1}{x+4}\right)$

**b** $\lim_{x \to -4^-} \left(\dfrac{1}{x+4}\right)$